



AMD Radeon RX 6000 Series

Check out the entire showcase presentation for AMD's latest graphics card lineup here. <https://digit.in/dec20-41>



Lisa Su dominates chip industry

AMD's CEO has received the semiconductor industry's highest honour - Robert N Noyce Award 2020. <https://digit.in/dec20-42>

AMD Radeon RX 6800 XT Graphics Card

Back in the game

RDNA 2 brings several new features to the table which have significant performance implications. The Compute Units have been changed to incorporate a new render pipeline which also include hardware-accelerated ray-tracing capabilities. There is also the new Smart Memory Access feature which helps the CPU address larger chunks of GPU memory. However, this is made possible because of the resizable BAR feature that's part of the PCIe specification and can be easily replicated by NVIDIA via software updates. So the competition is quite real and intense.

Specifications

The AMD Radeon RX 6000 series GPUs use the same 7nm process node so one wouldn't expect much power savings. AMD's architectural enhancements contribute to the rest of the performance gains. The key difference between the Radeon RX 6800 XT and the RX 6800 comes down to 12 Compute Units. RDNA 2 now has one Ray Accelerator per CU, so there should be a distinct difference in ray-tracing performance between the two cards.

Performance

The Radeon RX 6800 XT comes really close to the NVIDIA RTX 3090 in 3DMark Time Spy and Fire Strike Ultra. Considering that the latter is a \$1,499 card, this is a huge performance gain for a \$579 card. In BaseMark, the RX 6800 comes third, behind the RTX 3090 and the RTX 3080. What stood out for us was the OpenGL performance as the RDNA 2 cards tanked compared to the competition and are showing only a meagre generational improvement. In IndigoRender, the RTX 3090 and the RTX 3080 maintain a lead here whereas the RX 6800 XT and RX 6800 come in at the third and fourth positions.



In gaming benchmarks, the RX 6800 XT more or less equals the RTX 3080. There are some games where the former takes a lead and in some games, the latter is ahead. We saw the performance scale in a similar fashion as the NVIDIA games. Across several AAA games we saw 60 FPS+ at 4K resolution with all the bells and whistles. So it's a pretty good 4K gaming card.

Smart Access Memory

With a few tweaks in the BIOS on supported boards, you can switch on the Smart Access Memory feature and that's supposed to give the AMD processors a nice advantage over Intel processors. We tested the feature across four video games and found that it could go either way. Shadow of the Tomb Raider was the only game in which we saw AMD processors gain the advantage.

Pricing

The AMD Radeon RX 6800 is priced at ₹45,999 + Taxes and the pricing for the RX 6800 XT will be revealed later. With Taxes, the RX 6800 comes out to be ₹54,278 which is lesser than the RTX 3070 from NVIDIA but a teensy bit higher than the direct forex conversion. Since AMD has not revealed the local Indian pricing for the RX 6800 XT, we had to switch over to dollar



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pricing to get a better picture of the performance-per-dollar aspect. We've averaged the FPS values achieved by reference cards across 10 games and compared the same against their launch prices in the US. The surprising part is that higher-end GPUs are

giving much better value for money compared to the mid-range RX 5700 / RTX 2060 GPUs. This is not the norm since the sweet spot has always been with the mid-range GPUs and with AMD entering the fray in the top-end, even NVIDIA's prices, except the RTX 3090, have been decent. Both AMD and NVIDIA have comparable performance-per-dollar values with NVIDIA getting a slight lead.

Verdict

AMD has done it. The new 'RDNA 2'-based Radeon RX 6000 graphics cards are competitive, priced much lower and aren't power-hungry for the performance that they deliver. What remains to be seen is how the RX 6800 XT will be priced. Let's just hope that there's enough supply in the market.

—Mithun Mohandas

SPECIFICATIONS

STREAM PROCESSORS: 4608 | CU: 72 | BOOST CLOCK: 2250 MHz | VRAM: 16 GB GDDR6 | BUS-WIDTH: 256-bit | CONNECTORS: 2x 8-PIN | TBP: <300W

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